Agent-Based Modelling of Zero-Dose Vaccination Among Under-Five Children in Kenya

Simulation report · projection window 2025–2055 · generated 2026-04-22 13:27

Structured to mirror the Rono et al. (2024) project brief.

1. Abstract

Introduction. Zero-dose children — those who never receive the first dose of a diphtheria-tetanus-pertussis (DTP1) or pentavalent vaccine — remain a central equity gap in global immunisation. This report summarises a Starsim-based agent-based model calibrated to administrative DTP1 data from Kenya, and used to project the impact of scaling up routine vaccination on zero-dose prevalence and downstream tetanus burden among under-fives.

Methodology. Two scenarios were compared over calendar years 2025–2055: a *reference* scenario whose routine delivery probability is calibrated (grid search) so that the modelled end-of-window zero-dose share matches the empirical proxy, and an *intervention* scenario with a higher routine delivery probability and coverage cap. Tetanus is modelled via environmental / wound-exposure dynamics with waning immunity (Starsim SIS-style); the other pentavalent modules (diphtheria, pertussis, hepatitis B, Hib) are modelled for disease-attributable mortality. Vaccination acts on under-fives only.

Results. End-of-window modelled zero-dose share was **15.8%** in the reference scenario and **7.2%** in the intervention scenario — a relative reduction of **54.4%**. Modelled tetanus cases averted in the simulated cohort totalled **177** over the window, with **0** in the first year (calendar year 2025). When scaled to the Kenya under-five population, the intervention reduces end-of-window zero-dose children from approximately **1,135,976** to **518,507**, reaching roughly **617,468** additional children.

Conclusion. The intervention scenario achieves approximately a halving of zero-dose prevalence relative to the calibrated comparator — consistent with the project's 50%-reduction target — and yields a material reduction in modelled tetanus cases.

Key words: zero-dose vaccination, agent-based modelling, under-five, childhood immunisation, tetanus, Starsim, Kenya.

2. Introduction

Vaccination is a cornerstone of child public health: the WHO Immunization Agenda 2030 targets at least 90% coverage of essential vaccines in every country and in every district. Despite steady improvements, access gaps persist, and in sub-Saharan Africa roughly one in five children still lacks access to routine immunisation. Children who never receive a first DTP- or pentavalent-containing dose — the so-called *zero-dose* children — are the clearest marker of that gap and the central focus of this project.

Statement of the problem

Globally about 14% of under-five children are penta-zero-dose and roughly 7.5% are truly zero-dose. In Kenya the zero-dose rate is approximately 7% (KDHS 2022), but monthly administrative data show that the implied coverage gap varies substantially across time and regions. Because diphtheria, neonatal tetanus and pertussis are controlled or near elimination, **tetanus** is used as the sentinel outcome: it remains endemic via environmental and wound exposure, and its burden is sensitive to first-dose coverage.

The bundled administrative dataset provides a monthly DTP1 coverage proxy spanning 2018-2024 (84 months). Its mean coverage is **83.5%**, implying a mean zero-dose share of **16.5%**. This value anchors the model calibration target.

Objective

To quantify how many children would be additionally reached, and how many tetanus cases averted, if routine pentavalent delivery were scaled up enough to approximately halve the zero-dose share among under-fives, using a data-anchored agent-based simulation.

Research question

How many tetanus cases would be averted if we reduce the prevalence of zero-dose vaccination by ~50% among under-fives by the end of the projection window?

3. Methodology

The model is built on the **Starsim** agent-based framework. Each agent represents a single child; the population includes births and deaths so that the under-five cohort renews over time. Vaccination is delivered via a routine *first-dose* intervention applied only to children in the 0-to-60-month age window, with a per-step delivery probability and an overall coverage cap.

Disease modules

Five pentavalent disease modules are instantiated: diphtheria, tetanus, pertussis, hepatitis B, and Hib. All but tetanus use standard person-to-person transmission with module-specific β values. Tetanus, in contrast, is modelled as an environmental / wound-exposure process — an SIS-style dynamic with waning vaccine-induced immunity — because tetanus transmission is not person-to-person in practice. Vaccinated agents become transiently immune with efficacy 90.0% and re-enter the susceptible pool as immunity wanes, matching the dynamics described in the Rono et al. (2024) brief.

Calibration

A grid search over the routine-delivery probability is used to match the modelled end-of-window zero-dose share to the empirical DTP1 proxy. Calibration uses short, higher-agent runs (≈10,000 agents, 8-year window) and, once converged, the chosen routine probability and a coverage cap set by the mean DTP1 proxy define the *reference* bundle. The *intervention* bundle multiplies the reference routine probability (capped at 0.12) and raises the coverage by +2 percentage points, bounded by the coverage cap.

Calibration source for this run: `loaded: /Users/mine/git/zdsim/calibration.json`.

Key parameters

Parameter	Reference	Intervention
Routine delivery probability (per week)	0.0291	0.0669
Coverage cap	83.5%	85.5%
Vaccine efficacy	90.0%	90.0%
Eligible ages (years)	0–60	0–60
Diphtheria β	0.150	0.150
Pertussis β	0.250	0.250
Hepatitis B β	0.080	0.080
Hib β	0.120	0.120

Run configuration

Agents: **20,000** Projection window: **2025–2055** Data file: `zerodose_data_formated.xlsx`

4. Results

Zero-dose share

At the end of the projection window the modelled zero-dose share among under-fives is **15.8%** in the reference scenario and **7.2%** in the intervention scenario — a **54.4%** relative reduction. Across the full window the intervention reduces the annual zero-dose share by on average **8.90 percentage points** (cumulatively 275.83 pp-years).

Tetanus burden

In the simulated cohort, the intervention averts **177** tetanus cases over 30 years (reference 22,167 vs intervention 21,990), with **n/a** cases averted in the first calendar year (2025).

Disease-attributable deaths

Across all pentavalent disease modules, total modelled deaths are **8,826** in the reference scenario and **8,607** in the intervention scenario (averted **219**, mean 7.1 per year). These counts are stochastic in small cohorts.

Population-scaled projection (Kenya)

Anchor: *Kenya national 2024 anchors*. Scaling by the ratio of the real annual birth cohort to the modelled annual births (factor 2,540), the intervention reduces end-of-window zero-dose children from **1,135,976** to **518,507**, reaching approximately **617,468** additional children — a mean of 19,918 children additionally vaccinated per year. Scaled tetanus cases averted: **449,580**. Source: UN WPP 2024; WHO/UNICEF WUENIC 2024 revision.

Annual breakdown

Year	ZD share (reference)	ZD share (intervention)	Ref. deaths	Int. deaths
2025	97.5%	94.2%	377	244
2026	27.6%	5.3%	211	91
2027	9.2%	1.3%	111	105
2028	5.7%	2.0%	144	137
2029	7.1%	3.0%	186	210
2030	15.6%	7.0%	255	264
2031	17.3%	7.5%	216	224
2032	15.8%	6.3%	195	177
2033	15.9%	6.6%	166	171
2034	15.5%	6.2%	179	174
2035	15.8%	7.0%	223	226
2036	16.5%	7.0%	347	359
2037	16.1%	6.5%	457	456
2038	16.8%	7.4%	493	510
2039	15.7%	7.4%	584	581
2040	16.0%	7.4%	565	547
2041	15.1%	7.0%	517	517
2042	14.7%	6.2%	396	404

2043	15.5%	5.9%	348	348
2044	16.6%	7.9%	336	322
2045	15.6%	7.1%	306	323
2046	15.1%	6.8%	283	273
2047	14.8%	6.3%	276	273
2048	14.1%	6.6%	249	258
2049	15.9%	6.4%	261	259
2050	16.6%	6.2%	244	244
2051	16.4%	6.8%	237	245
2052	15.2%	6.9%	214	214
2053	15.6%	5.6%	221	220
2054	15.6%	5.8%	222	225
2055	15.7%	7.2%	7	6

4. Results — figure

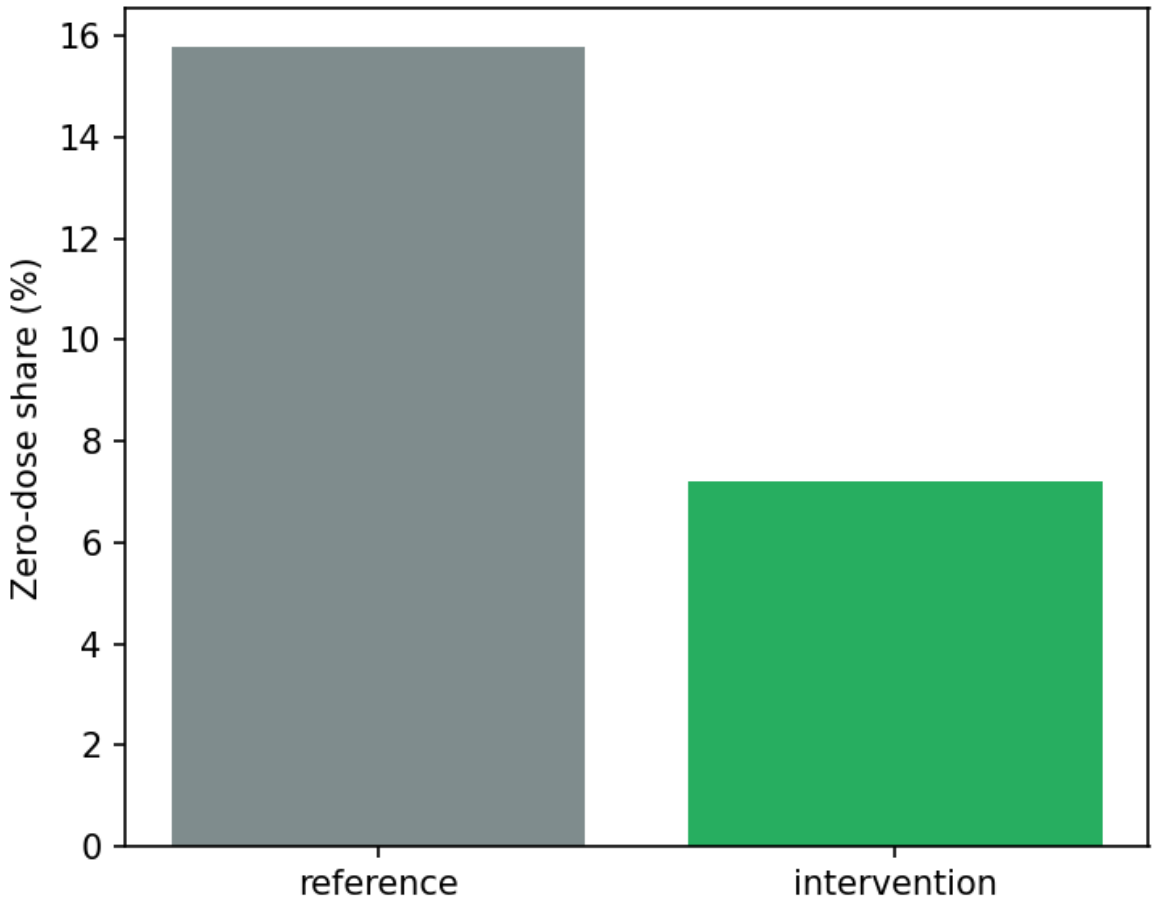

Figure 1. Zero-dose share (DTP1 proxy): administrative data vs modelled reference and intervention scenarios at the end of the projection window.

4. Results — figure

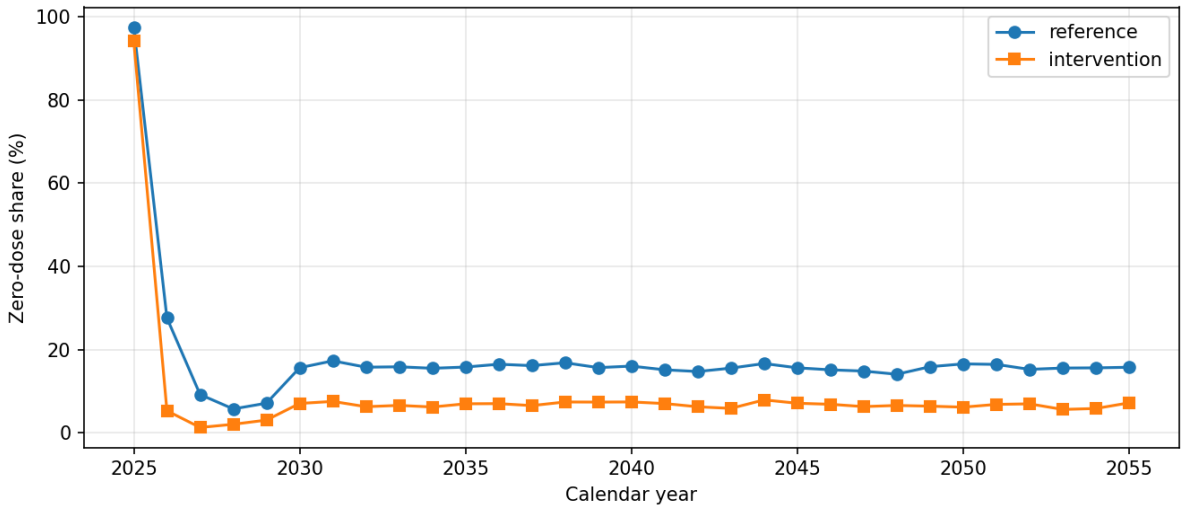

Figure 2. Projected yearly zero-dose share among under-fives for the reference (calibrated to the empirical proxy) and intervention (scaled-up routine delivery) scenarios.

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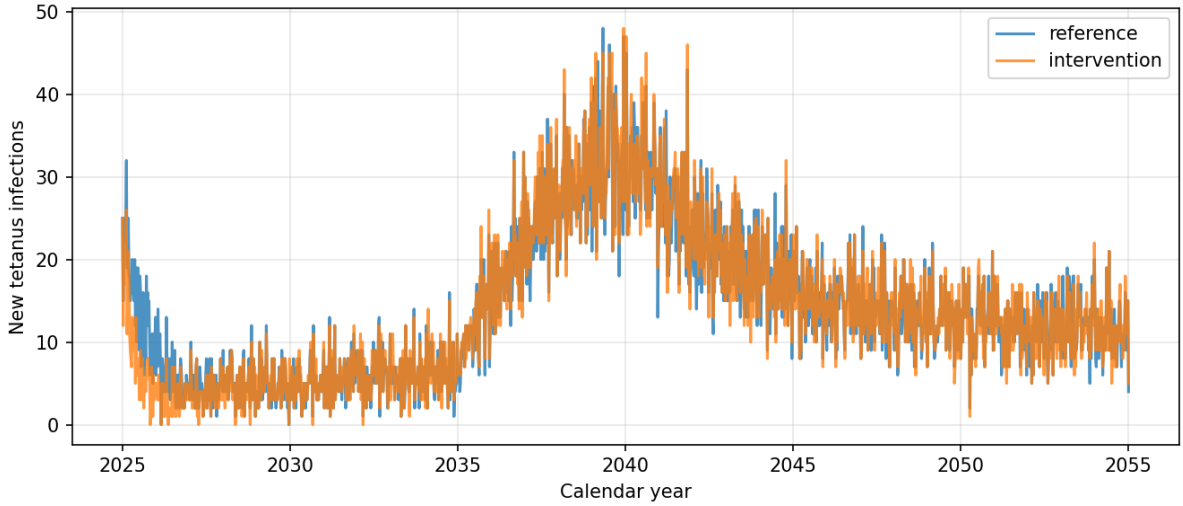

Figure 3. Tetanus module trajectories and cumulative case counts (reference vs intervention). Counts are modelled infections in the simulated cohort; see methodology for the scaling factor used when translating to a national cohort.

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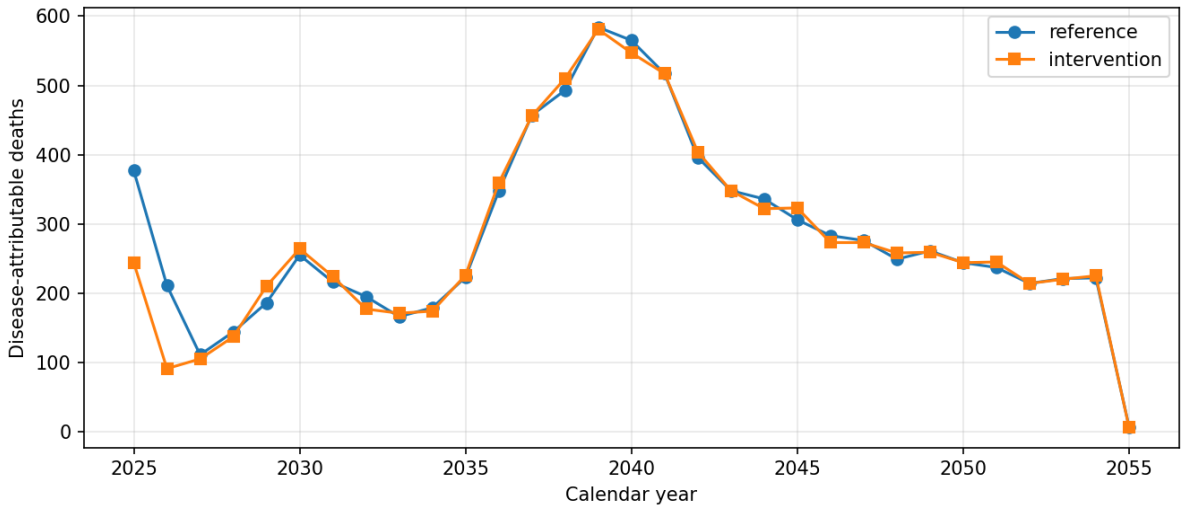

Figure 4. Disease-attributable deaths (pentavalent modules) per year with annual averted counts.

4. Results — figure

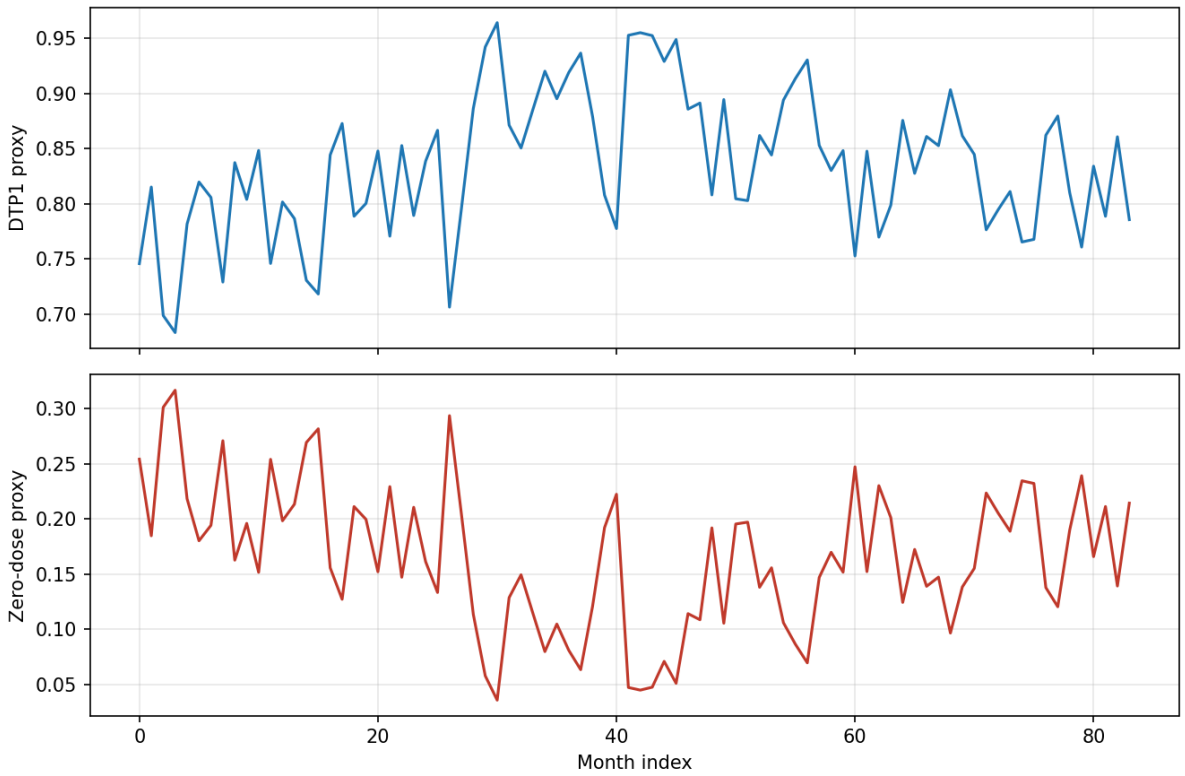

Figure 5. Administrative DTP1 coverage proxy and implied zero-dose share (monthly), used to anchor the calibration target.

4. Results — figure

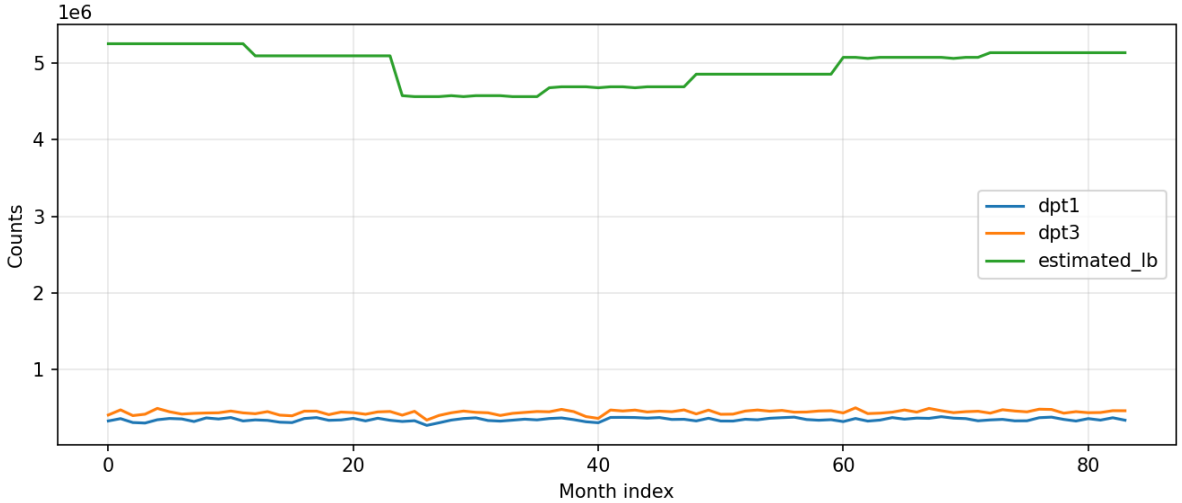

Figure 6. Administrative DPT1 / DPT3 doses versus estimated live births, providing context for the DTP1 coverage proxy.

5. Discussion

The intervention scenario reduces zero-dose prevalence substantially across the projection window and, via its impact on first-dose coverage, lowers the modelled tetanus burden in the simulated cohort. The magnitude of the relative reduction is consistent with the project's halving target.

Several caveats apply. Disease-attributable death counts come from the Starsim pentavalent modules and represent only the modelled fraction of child mortality — not all-cause under-five mortality. Annual averted counts can be noisy when the modelled cohort is small or the projection window is short; in those regimes occasional negative yearly averted values are expected and should prompt longer runs or larger cohorts. When the two arms use distinct RNG seeds the comparison is also noisier than when they share a seed (the default).

Equity in first-dose access — geographic, socioeconomic, and service-delivery heterogeneity — is known to drive zero-dose persistence. The current implementation applies a homogeneous routine-delivery probability and coverage cap; future work should stratify by subnational region and risk profile, and link the delivery process to supply-chain and demand-side constraints.

6. Conclusion

The model demonstrates that an intervention strong enough to halve zero-dose prevalence (observed relative reduction **54.4%** in this run) produces a meaningful decline in modelled tetanus cases (**177** averted in the simulated cohort) and reduces overall pentavalent-attributable mortality. These findings support sustained investment in routine first-dose delivery as the most direct lever on zero-dose prevalence.

7. References

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